

1 ASIO Full Analysis Report

This report summarizes the analysis of dataset 20260604.0x02B4. The trial spans 2256.49 s of continuous data acquisition across all six channels. The remainder of this section reports both raw and detrended + EMI-filtered statistics. Units are femtoamps for mean/RMS/std; skew and kurtosis are dimensionless.

1.1 Detrended + EMI-filtered Statistics

Table 1: Detrended + EMI-filtered Current Statistics

Channel	Mean (fA)	RMS (fA)	Std Dev (fA)	Skew	Kurtosis
SXR1	1.3329×10^{-16}	4.6884×10^1	4.6884×10^1	-1.9938×10^{-3}	8.9090×10^{-2}
SXR2	7.1697×10^{-16}	4.6863×10^1	4.6863×10^1	3.3327×10^{-2}	1.4417×10^{-2}
SXR3	4.8249×10^{-16}	5.3991×10^2	5.3991×10^2	3.8089×10^2	1.6054×10^5
SXR4	4.7961×10^{-16}	4.4730×10^1	4.4730×10^1	1.2535×10^{-1}	2.1595×10^{-2}
HXR	2.1851×10^{-14}	7.8835×10^2	7.8835×10^2	1.4741×10^{-2}	4.6225×10^{-2}
EUV	-1.5888×10^{-14}	7.2072×10^2	7.2072×10^2	5.7140×10^{-3}	5.0090×10^{-2}

1.2 Raw Statistics

Table 2: Raw Current Statistics

Channel	Mean (fA)	RMS (fA)	Std Dev (fA)	Skew	Kurtosis
SXR1	5.2328×10^4	5.2328×10^4	6.9331×10^1	-1.2443×10^{-1}	-1.3340×10^{-1}
SXR2	5.4434×10^4	5.4434×10^4	5.4695×10^1	-1.4859×10^{-2}	-1.8411×10^{-2}
SXR3	5.5039×10^4	5.5042×10^4	5.5832×10^2	4.0437×10^2	1.7360×10^5
SXR4	5.2913×10^4	5.2913×10^4	6.0599×10^1	-3.7087×10^{-2}	6.9656×10^{-2}
HXR	6.7849×10^5	6.7849×10^5	2.0209×10^3	3.3129×10^{-1}	-7.6470×10^{-1}
EUV	7.5482×10^5	7.5482×10^5	1.0072×10^3	-3.4448×10^{-1}	1.2862×10^{-1}

1.3 Trial Duration

Quantity	Value
Total acquisition time	2256.490 s

1.4 Raw Voltage Time Series

ASIO Raw Voltage Values

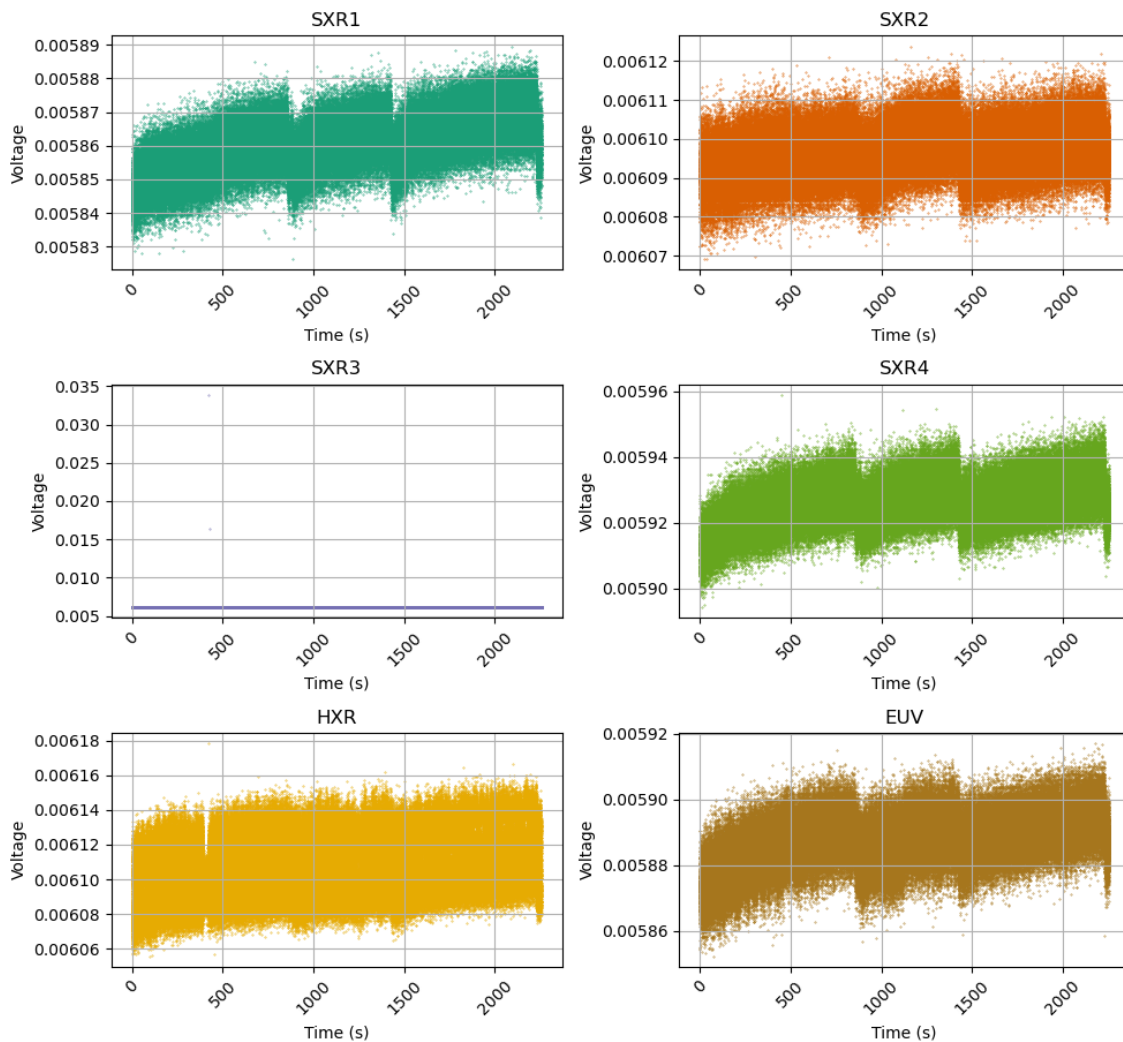


Figure 1: Raw voltage vs. time (s) for all six ASIO channels.

1.5 FFTs of Raw Voltages

FFTs

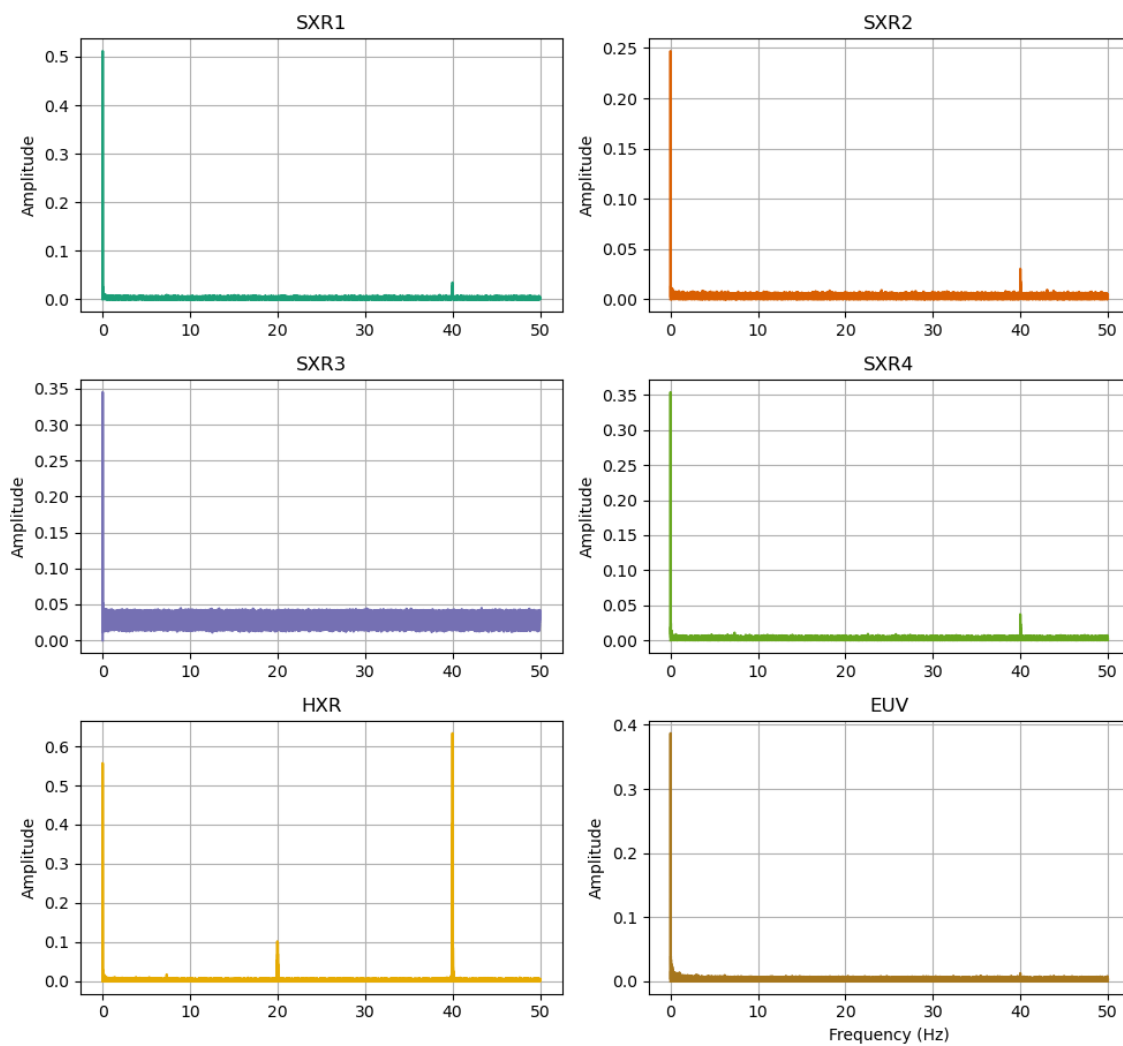


Figure 2: FFTs of raw voltages (mean centered).

1.6 Detrended + EMI-filtered Time Series

EMI-Filtered and Detrended Voltage Values

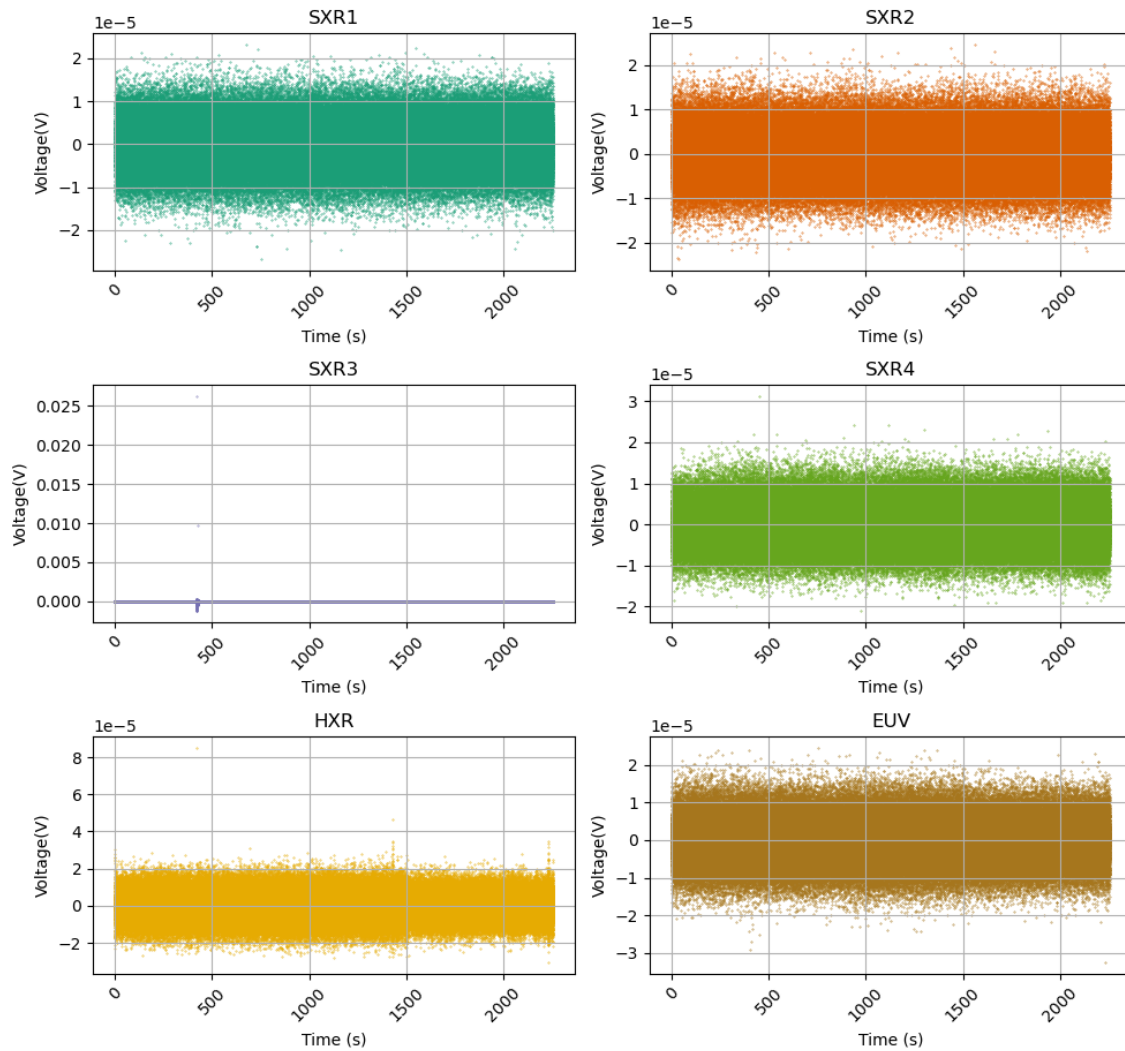


Figure 3: Voltage vs. time (s) after Savitzky–Golay detrending and 40 Hz EMI removal.

1.7 Noise Histograms

ASIO Noise Histograms w/EMI-filtering and Detrending

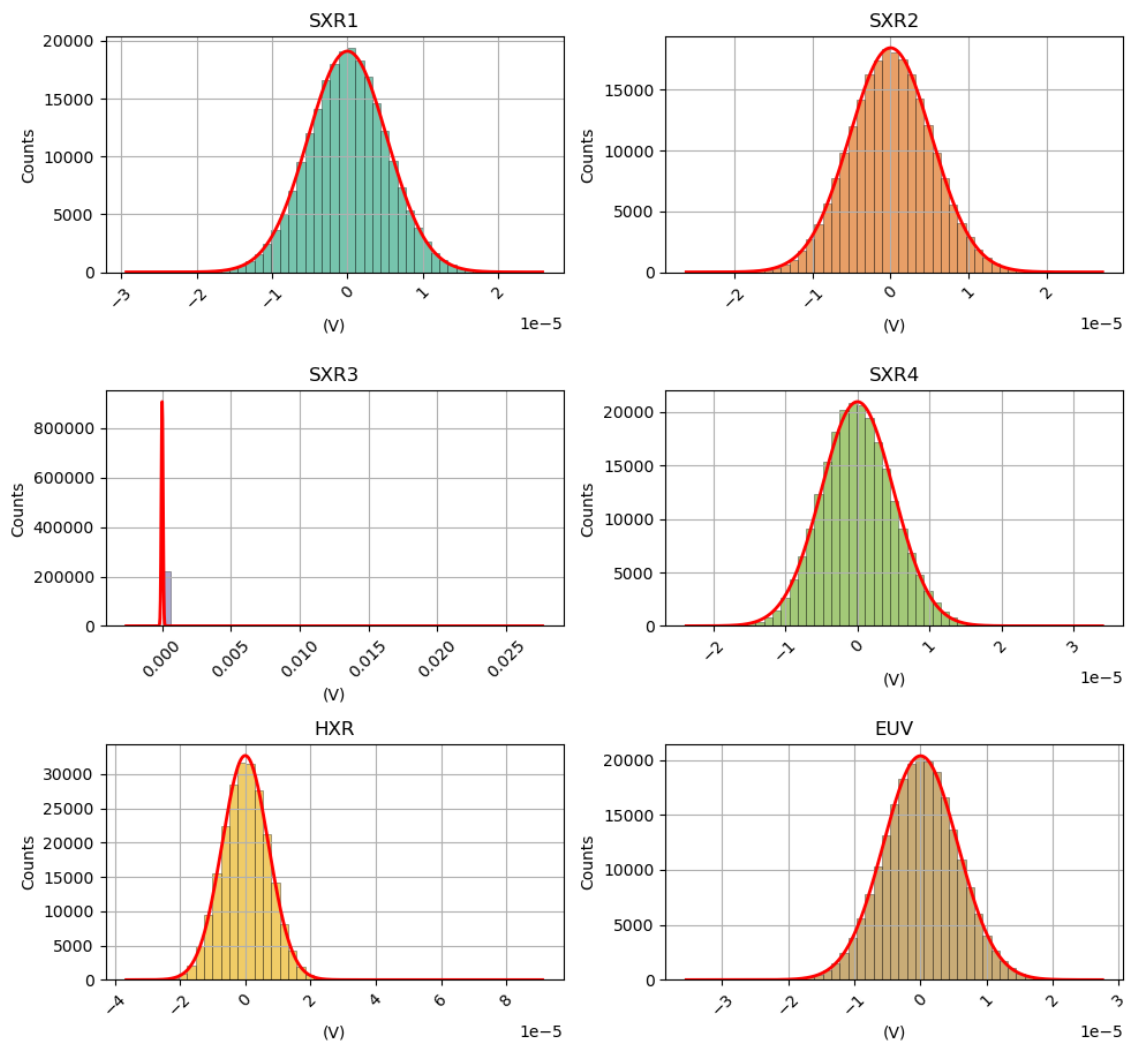


Figure 4: Noise histograms after detrending and 40 Hz EMI removal. Red curve is a Gaussian fit.

1.8 Fe-55 Expectation Values

Fe-55 expectation values: not yet implemented.

Testing Notes

selection: whole dataset (t=0.0s–2256.5s)